Kareem Abu Salah

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Education

Duke University, Pratt School of Engineering Durham, NC

B.S. Mechanical Engineering, A.B. Computer Science *Expected May 2028*

- **GPA: 3.56** | Junior | Coursework: Mechanics of Solids, Statics, Product Design, Linear Algebra, Probability

Work Experience

Embedded & Mechanical Intern | FaunaLabs *(marine sensor-hardware startup; whale tags)* Mar–Sep 2026

- Qualified a sealed electronics enclosure to 20 MPa (2 km) with a proof pass and 1,000+ seal fatigue cycles.
- Cut dropped sensor frames 90% by fixing a Cortex-M7 D-cache/DMA race with a non-cacheable MPU region.
- Shrank tag part count 30% by consolidating the seal, cutting 4 minutes from per-unit assembly.

Test & Battery Engineering Intern | Jordan Int'l Air Cargo & Defense May–Aug 2025

- Accelerated Li-ion pack qualification 40% by automating discharge testing on a custom-built test stand.
- Reduced downstream scrap 15% by troubleshooting 30+ failed Li-ion packs to root cause, e.g., a weak spot-weld.
- Delivered 15 flight-ready fixed-wing UAVs in one summer, integrating and checking out airframe, power, and avionics.

Drone Lab Intern | HTU AeroSquad May–Jul 2024

- Damped gyro oscillation 30% across 26 FPV builds by tuning P and D gains in Betaflight per airframe.
- Built and soldered power distribution boards, speed controllers, and LiPo packs across 26 FPV drones.
- Standardized safety training for 20+ new members by authoring a department-wide written SOP.

Project Experience

Founder & Hardware/Firmware Lead | Plantogachi *(consumer sensor device + app)* May 2026–Present

- Kept the BOM at $32 ($11 quoted at bulk) with a tool-free snap-fit enclosure of uniform 2 mm walls.

Undergraduate Researcher | Duke Cyber-Physical Systems Lab *(drone & robot research platforms)* Mar–Sep 2026

- Trimmed a hexacopter propeller guard's drag 36% in Ansys Fluent CFD by reprofiling its cross-section.
- Laid out a drone's 5/12/19 V power-distribution PCB in Altium, checking the layout for RF interference.
- Ran 30+ test flights on three Aurelia X-600 hexacopters as the lab's only trained operator.
- Time-synced radar, lidar, and depth cameras on three robot platforms over ROS 2 for sensor fusion.

Structures and Integration & Test Lead | Duke Solar Sail Spacecraft *(Earth Cup competition)* Mar 2026–Present

- Predicted a 27.6 Hz first mode in 402-DOF Python FEA, hand-checked within 4.3%, above the standard's 20 Hz floor.
- Landed the frame 25% under mass budget, 2.96 g per beam, sizing 8.0/0.15 mm tubes in a seven-geometry sweep.
- Defined an 8-cycle, −40 to +85 °C thermal-vacuum test plan to catch creep at the carbon-to-nylon bond.
- Writing the acceptance procedure for the team's in-house magnetorquer, verifying polarity on a water-float 0 g jig.

Designer | McRocket *(injection-molded product)* Feb 2026

- Stacked ±0.002" tolerances worst-case into a 0.250"/0.254" snap-fit, measuring 0.004" on the prototype.

Repass-Rodgers Scholar | Duke University Marine Lab Dec 2025–Present

- Won a $5,500 grant (1 of 4 selected) to test whether low-cost sensors can match expensive oceanographic gear.

Structures & Flight-Computer Engineer | Duke SEDS *(rocketry & CubeSat team)* Sep 2025–Present

- Flew a CubeSat to near space (28 km) by balloon, recovering its GPS altitude profile for Duke researchers.
- Analyzed a 1U CubeSat chassis in FEA to 2× margin at the 9 g CubeSat launch-load spec.
- Simplified CubeSat chassis assembly to 8 fasteners by keying 6 jigsaw panels that clamp top-to-bottom.

Hardware Engineer | Duke Engineering Design *(tiger enrichment dispenser, NC Zoo client)* Aug–Dec 2025

- Removed 67% of keepers' daily manual labor with an Arduino solenoid dispenser dosing 6 mL oil every 5 hours.

Skills

Mechanical Design & Analysis: SolidWorks, Onshape, CAD modeling & high-quality engineering drawings, GD&T, DFM/DFA, tolerance stack-up, FEA (Python, Ansys), CFD (Ansys Fluent), first-principles hand calculations

Fabrication & Build: CNC machining (mill & lathe, G-code), 3D printing (FDM, SLS), laser cutting, CFRP wet layup, PCB assembly & soldering, hands-on prototyping; Pratt Student Shop Safety Certified

Test & Manufacturing: integration & test of flight hardware, test plans, custom test stands & fixtures, thermal-vacuum & modal test planning, pressure & fatigue testing, root-cause analysis, scrap reduction, design reviews

Electrical & Software: embedded C/C++, Python, MATLAB, STM32, ESP32, FreeRTOS, ROS 2, Altium, LTspice, Git; PID control tuning, spacecraft subsystems integration across mechanical, electrical & software

Additional: collegiate CubeSat & rocketry design/build team; FAA Part 107 Remote Pilot; U.S. permanent resident (U.S. Person under EAR, green card, no sponsorship needed)